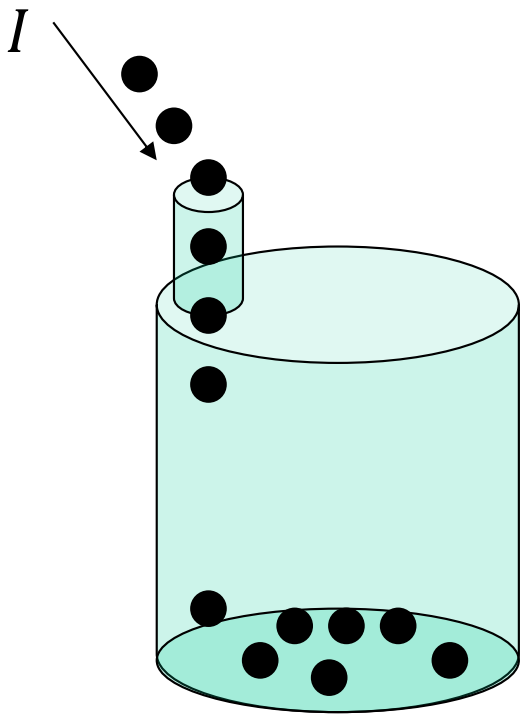
ECE 329
Lecture 16

Conservation of Charge
Boundary Conditions

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Conservation of Charge



Say we have a container that can accumulate charge

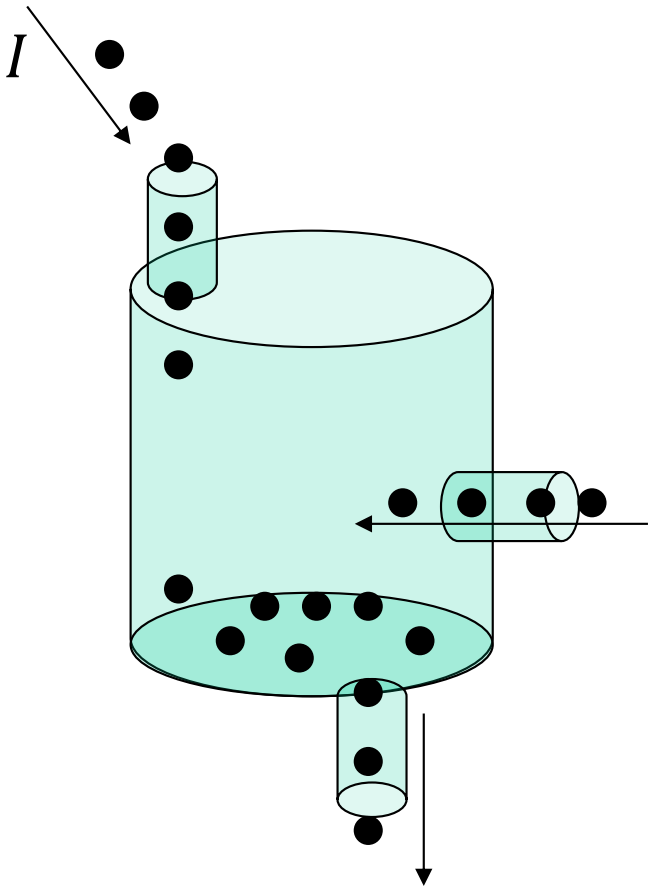
Start pouring charges into the container

Flow of charges is a current, I, in Amps

If the charges don't leave the container the charge inside the container increases

Current flow IN = Charge INCREASE
Current flow OUT = Charge DECREASE

Conservation of Charge



In general, we can pour charges in from more than one direction, or take some out from other parts of the container

Net Rate of Current flow OUT = Net Rate of Charge DECREASE

$$I_{out} = \oiint_S \vec{J} \bullet d\vec{S} = -\frac{dQ_{enc}}{dt} = -\frac{d}{dt}\iiint_V \rho dV$$

$$= \iiint_V \nabla \cdot \vec{J}\, dV$$

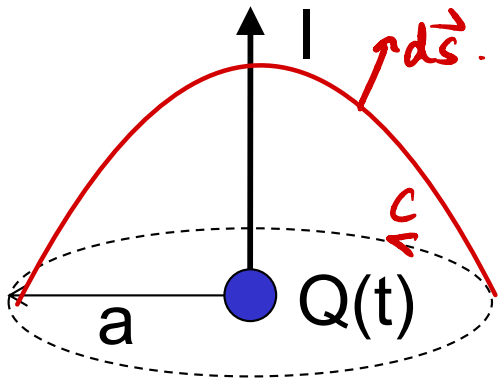
or, $$\nabla \cdot \vec{J} = -\frac{d\rho}{dt}$$

using the divergence theorem

Continuity equation

Application of multiple Maxwell's Equations

- Current I flows from a point charge Q(t) at the origin along the z-axis off to infinity. Find the counterclockwise MMF for a circular path of radius a in the xy plane centered at the origin. Hint: consider a sphere and a hemisphere



semisphere

$$MMF = \oint_C \vec{H}\cdot d\vec{l} = I + \frac{d}{dt}\iint_S \vec{D}\cdot d\vec{s}$$

$\iint_S \vec{J}\cdot d\vec{s}$ $\quad = \frac{Q(t)}{2}$

$$\oiint_S \vec{D}\cdot d\vec{s} = Q_{enc}$$

whole sphere

$$= I + \frac{d}{dt}\left(\frac{Q(t)}{2}\right)$$

$$I = \iint \vec{J}\cdot d\vec{s} = -\frac{d}{dt}Q(t)$$ charge conservation

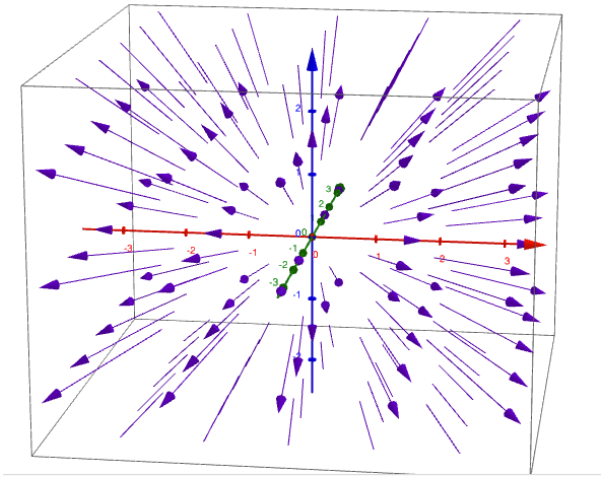
$$= I - \frac{I}{2} = \boxed{I/2}$$

From Example 2.5 (p 111) of old book

Application of charge conservation

- For **J**=<x,y,z>, find the rate of decrease of charge contained in the unit cube: corner vertices (0,0,0) and (1,1,1).

$-\frac{d}{dt}Q(t)$ = 3A.



$$\oint_S \vec{J}\cdot d\vec{S} = -\frac{d}{dt}Q(t)$$

↓ Div. theorem.

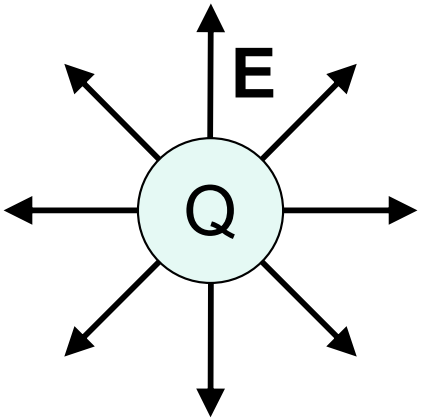
$$\iiint_V \nabla\cdot\vec{J}\, dV$$

$$= \iiint_V \left(\frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}\right)\cdot\left(x\hat{x}, y\hat{y}, z\hat{z}\right) dV.$$

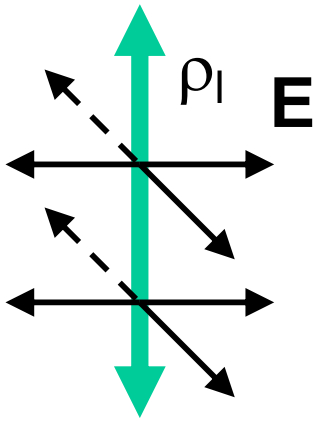
$$= \iiint_V 3\, dV = 3\ [A].$$

From Problem 2.24a (p 126) of old book

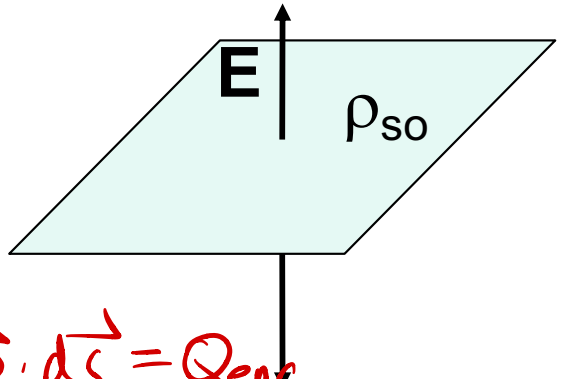
E and **B** for basic configurations



$$\vec{E} = \frac{Q}{4\pi\varepsilon_0 R^2}\hat{a}_R$$

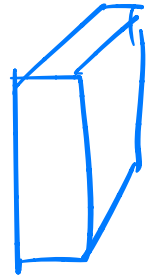


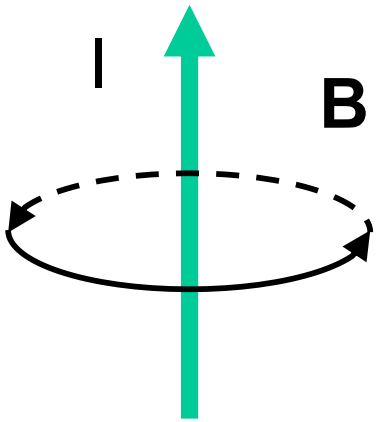
$$\vec{E} = \frac{\rho_l}{2\pi\varepsilon_0 r}\hat{a}_r$$



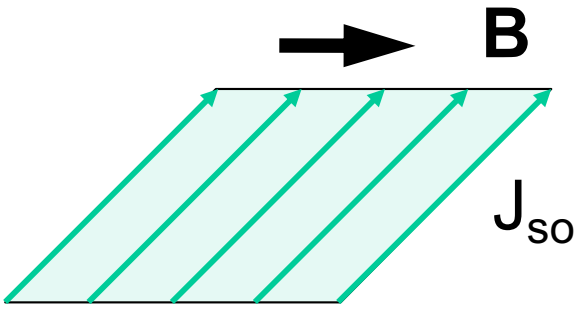
$$\oint_S \vec{D}\cdot d\vec{s} = Q_{enc}$$

$$\vec{E} = \frac{\rho_{so}}{2\varepsilon_0}\left(\pm\hat{a}_z\right)$$





$$\vec{B}_\phi = \frac{\mu_0 I}{2\pi R}$$



$$\oint_C \vec{H}\cdot d\vec{l} = I_{enc}$$

$$B = \frac{\mu_0}{2}\vec{J}_s \times \hat{a}_n$$



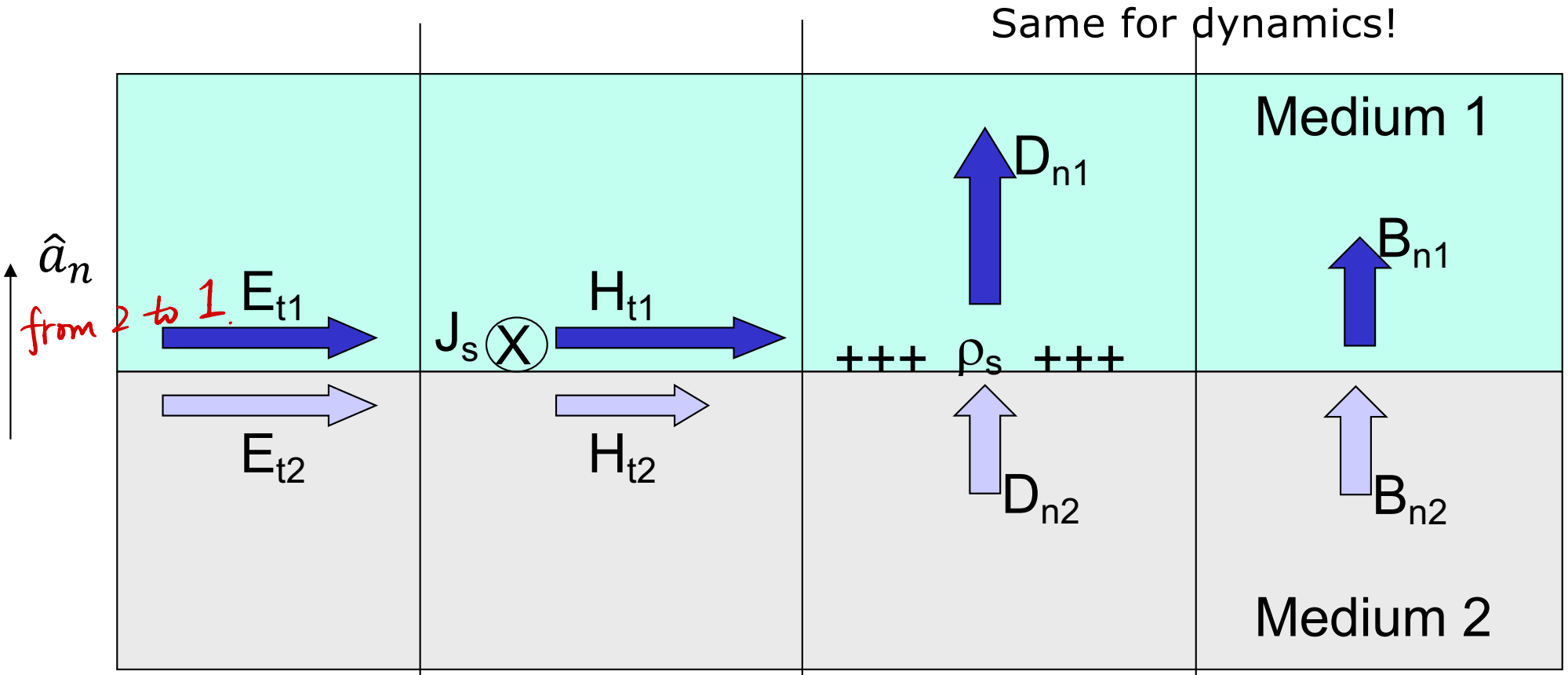


Lecture 16a Summary

- Maxwell's Equations
  - Gauss Magnetic: _____
  - Gauss Electric: _____
  - Faraday: _____
  - Ampere + Maxwell:

    _____

Boundary Conditions



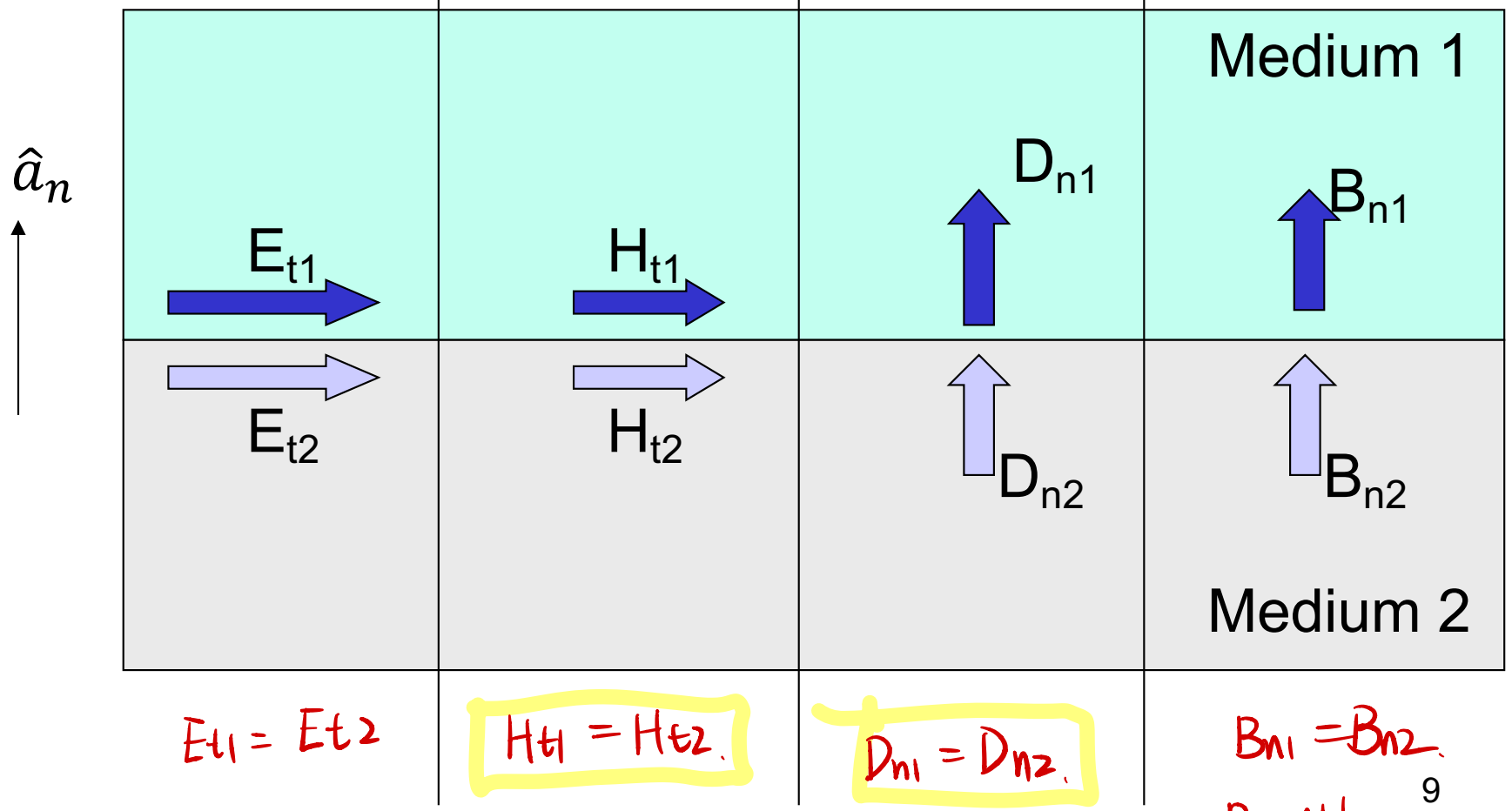
$$E_{t1} - E_{t2} = 0 \qquad |\vec{H}_{t1}| - |\vec{H}_{t2}| = \pm|\vec{J}_s| \qquad D_{n1} - D_{n2} = \rho_s \qquad B_{n1} - B_{n2} = 0$$

$$\hat{a}_n \times (\vec{E}_1 - \vec{E}_2) = 0 \qquad \hat{a}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s \qquad \hat{a}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s \qquad \hat{a}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0$$

Boundary between Two Perfect Dielectrics

$$\sigma = 0 \qquad J_S = \sigma E = 0 \qquad \rho_S = 0$$

no free charge.



$E_{t1} = E_{t2}$

$H_{t1} = H_{t2}$.

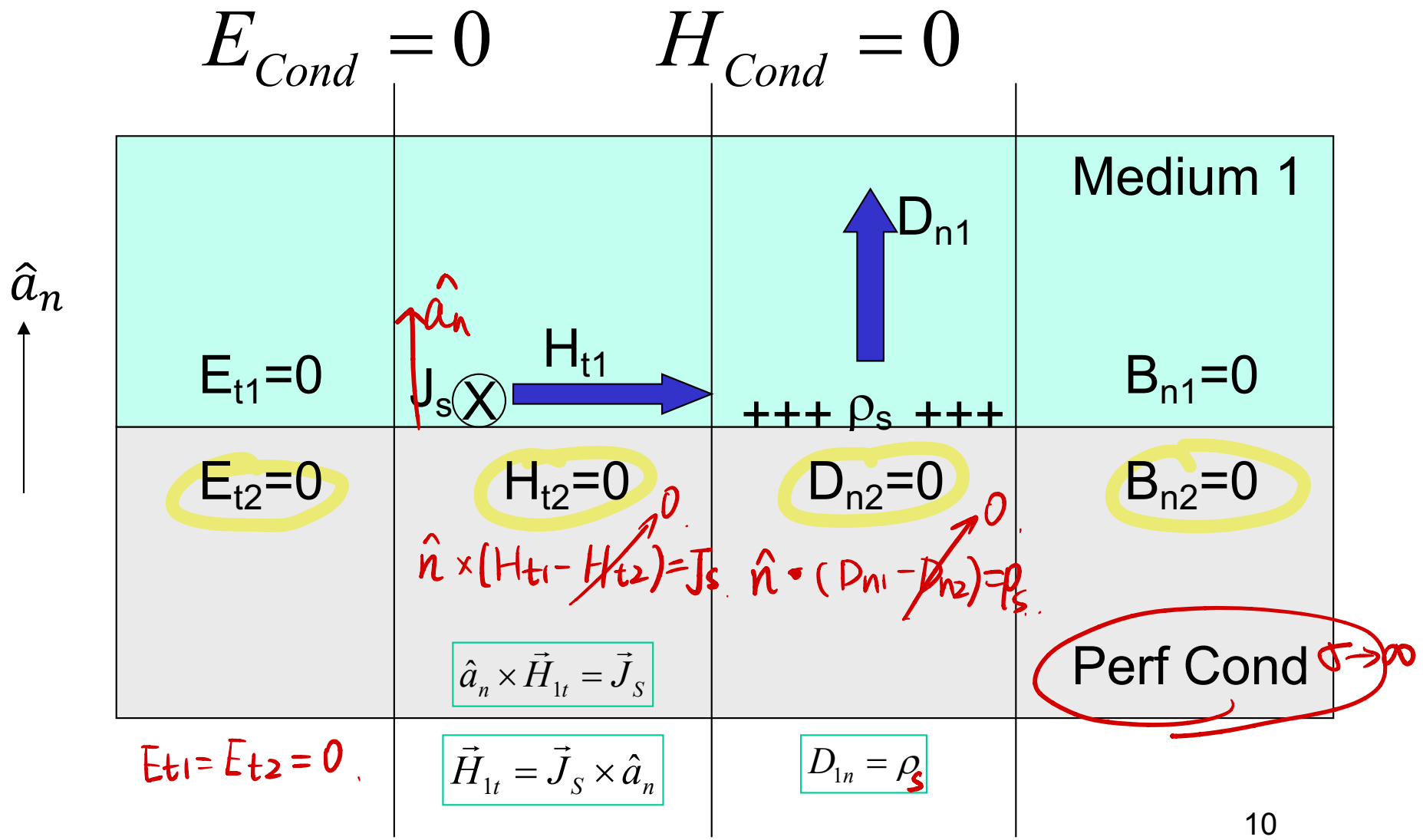
$D_{n1} = D_{n2}$.

$B_{n1} = B_{n2}$.

$B = \mu H$

$\mu_1 H_{n1} = \mu_2 H_{n2}$.

Surface of Perfect Conductor

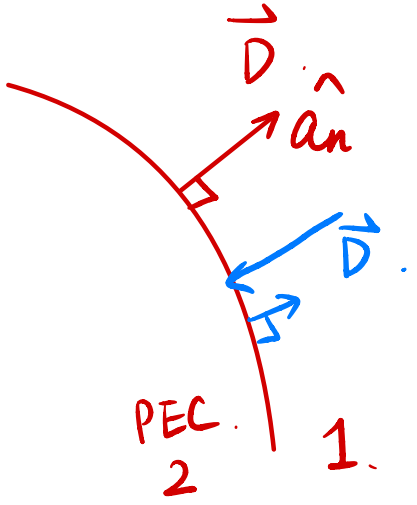


Finding surface charge

- Given **D** on the surface of a perfect conductor, find the surface charge, ρ_s, at that point (Hint: what is the vector $\mathbf{a}_n$ for each surface):

(a) $\mathbf{D}=D_0\langle 1,-2,2\rangle$ pointing away from surface

(b) $\mathbf{D}=D_0\langle 1,0,\sqrt{3}\rangle$ pointing towards surface



$\hat{a}_n \cdot \vec{D} = \rho_s$ $\quad E_{t1} = 0$

$\frac{\vec{D}}{|D|} \cdot \vec{D} = \rho_s = |D|$

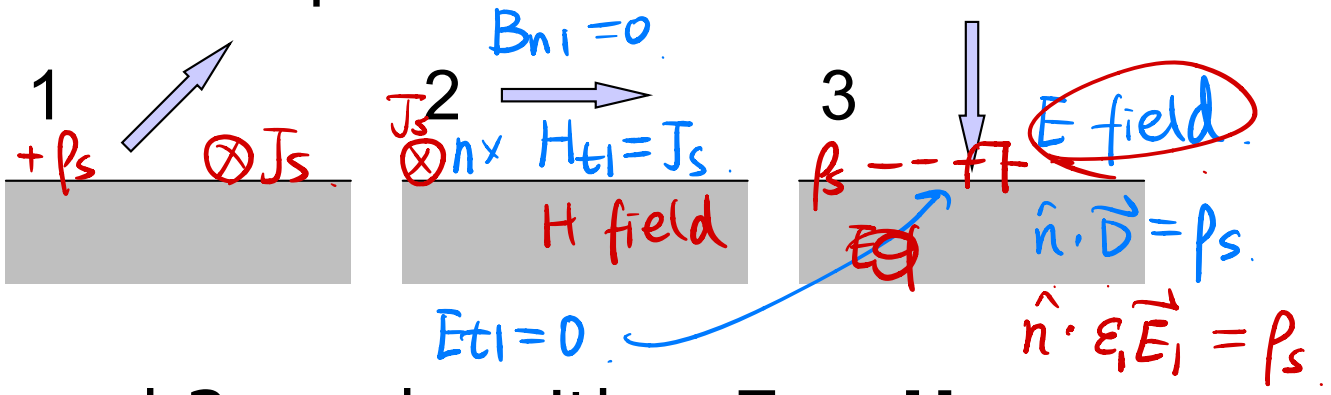
(a) $\rho_s = |D| = D_0\sqrt{1^2+(-2)^2+2^2} = 3D_0 > 0$

(b) $\rho_s = -|D| = -D_0\sqrt{1+3} = -2D_0$ $[C/m^2]$

From D4.13 (p262) of old book

Challenge Question: Boundary conditions

- Which of the following are realizable as the field outside a perfect conductor



(a) 1, 2, and 3 can be either **E** or **H**

(b) 1 can be either **E** or **H**, but 2 is **E**, 3 is **H**

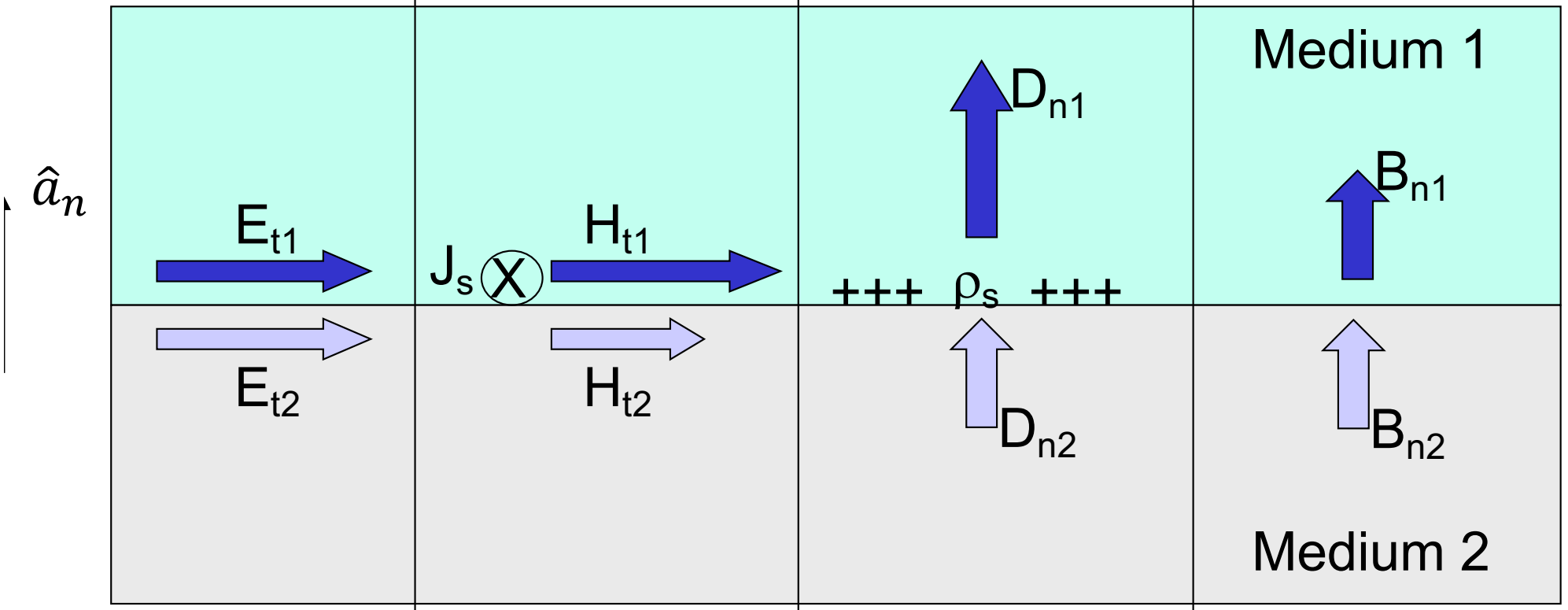
(c) 1 can be either **E** or **H**, but 2 is **H**, 3 is **E**

(d) 1 can be neither **E** nor **H**, but 2 is **E**, 3 is **H**

(e) 1 can be neither **E** nor **H**, but 2 is **H**, 3 is **E**

Lecture 16b Summary

- **Never** use the differential form of Maxwell's equations at a boundary – only use integral form



$$E_{t1} - E_{t2} = 0 \qquad |\vec{H}_{t1}| - |\vec{H}_{t2}| = \pm|\vec{J}_s| \qquad D_{n1} - D_{n2} = \rho_s \qquad B_{n1} - B_{n2} = 0$$

$$\hat{a}_n \times (\vec{E}_1 - \vec{E}_2) = 0 \qquad \hat{a}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s \qquad \hat{a}_n \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s \qquad \hat{a}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0$$